Jingzhe Shi

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Work Experience:

Trading Intern — Optiver, Shanghai Office, China (July 2024 – August 2024)

- Optiver is a leading global market-making firm headquartered in the Netherlands, with offices in financial centers worldwide. Shanghai is a very important office.
- Completed a competitive summer internship in the trading division, gaining hands-on exposure to electronic market making, options pricing, and quantitative trading strategies.
- Participated in trading simulations, daily market analysis, and team-based problem-solving exercises designed to evaluate decision-making under uncertainty and time pressure.

Education:

Tsinghua University — Beijing, China (September 2021 – June 2025)

- Bachelor of Engineering, Yao Class (an honors program in computer science), Institute for Interdisciplinary Information Sciences.
- Grade Point Average: 3.86 / 4.0 (overall); 3.90 / 4.0 (specialized courses).
- **Relevant Coursework (all earned a 4.0 / 4.0 grade)**: Machine Learning; Natural Language Processing; Advanced Computer Graphics; Design Thinking in the Age of Artificial Intelligence; Computer Architecture; Operating Systems and Distributed Systems; Mathematics for Computer Science; Theory of Computation; The Physics of Information; and Research Immersion Training.
- **Undergraduate Research Activities**: Conducted multiple research projects in machine learning, large language models, and time series analysis under faculty supervision at Tsinghua University, resulting in several first-authored and co-authored publications at top-tier international machine learning conferences (full list provided in the Publications section below).

University of California, San Diego (UCSD) — California, USA (Feb 2024 – June 2024)

- Exchange research internship. (Visa type: J1)
- Focus on research on time series and agent systems (full list provided in the Publications section below).

The Second High School Attached to East China Normal University — Shanghai, China (graduated June 2021)

- High school diploma. Won gold medal in International Physics Olympiad.

Publications and Patents:

The asterisk (*) denotes equal contribution; the hash symbol (#) denotes equal corresponding authorship.

Intrinsic Entropy of Context Length Scaling in Large Language Models — Jingzhe Shi*, Qinwei Ma*, Hongyi Liu*, Hang Zhao#, Jeng-Neng Hwang, and Lei Li#. Accepted to the International Conference on Learning Representations (ICLR) 2026. May 2025.

- Studies how long contexts have impact on language models.
- OpenReview link: https://openreview.net/forum?id=vnipyA8c9V.

PRISM-Physics: Causal Directed Acyclic Graph Based Process Evaluation for Physics Reasoning — Wanjia Zhao*, Qinwei Ma*, Jingzhe Shi*, Shirley Wu, Jiaqi Han, Yijia Xiao, Si-Yuan Chen, Xiao Luo, Ludwig Schmidt, and James Zou. Accepted to the International Conference on Learning Representations (ICLR) 2026. October 2025.

- Studies how to benchmark language models with physics olympiad problems and good process reward.
- OpenReview link: https://openreview.net/forum?id=4PZMeopXzP.

Physics Supernova: An Artificial Intelligence Agent that Matches Elite Gold Medalists at the International Physics Olympiad 2025 — Jiahao Qiu*, Jingzhe Shi*, Xinzhe Juan, Zelin Zhao, Jiayi Geng, Shilong Liu, Hongru Wang, Sanfeng Wu, and Mengdi Wang. Accepted to the NeurIPS 2025 Workshop on Evaluating Large Language Models (oral presentation) and to the NeurIPS 2025 Workshop on Large Foundation Models for Educational Assessment (spotlight presentation). September 2025.

- Studies how to use multi-agent system to solve physics olympiad problems, like IPhO 2025.
- ArXiv link: https://arxiv.org/abs/2509.01659.

Scaling Law for Time Series Forecasting — Jingzhe Shi*, Qinwei Ma*, Huan Ma, and Lei Li. Accepted to the Conference on Neural Information Processing Systems (NeurIPS) 2024. May 2024.

- Studies how time series forecasting neural networks behave for longer horizon, larger training dataset, and larger model.
- OpenReview link: https://openreview.net/forum?id=Cr2jEHJB9q.

Graph Canvas for Controllable 3D Scene Generation — Libin Liu, Shen Chen, Sen Jia, Jingzhe Shi, Zongyu Jiang, Can Jin, Wu Zongkai, Jeng-Neng Hwang, Lei Li. Accepted to the ACM International Conference on Multimedia (ACM MM) 2024. Nov 2024.

- Studies how to use multi-agent system to help generate 3D scenes.
- ACM link: https://dl.acm.org/doi/abs/10.1145/3746027.3754554.

CHOPS: Chat with Customer Profile Systems for Customer Service with Large Language Models — Jingzhe Shi, Jialuo Li, Qinwei Ma, Zaiwen Yang, Huan Ma, and Lei Li. Accepted to the Conference on Language Modeling (COLM) 2024. April 2024.

- Studies how to use multi-agent system to help with customer service tasks.
- OpenReview link: https://openreview.net/forum?id=9Wmdk94oKF.

Large Trajectory Models are Scalable Motion Predictors and Planners — Qiao Sun, Shiduo Zhang, Danjiao Ma, Jingzhe Shi, Derun Li, Simian Luo, Yu Wang, Ningyi Xu, Guangzhi Cao, and Hang Zhao. Preprint, October 2023.

- Studies how to use large neural networks to solve trajectory prediction and planning tasks for autonomous driving scenarios.
- ArXiv link: https://arxiv.org/abs/2310.19620.

No patents held.

Professional Certifications:

Test of English as a Foreign Language (TOEFL iBT) — Educational Testing Service (ETS), United States. Test taken in November 2024.

- Overall score: 113 out of 120 (Reading: 30; Listening: 28; Speaking: 26; Writing: 29).

Awards, Interests, Skills, and Community Involvement:

Awards and Honors:

- **Gold Medal, 51st International Physics Olympiad (IPhO 2021)** — July 2021.
  International Physics Olympiad is an international Olympiad for high school students to attend with problems featuring interesting physics phenomena.

Interests:

- Research interests include machine learning, large language models, scaling laws and theoretical foundations of deep learning, reasoning in artificial intelligence systems, and applications of artificial intelligence to physics.
- Personal interests include physics competitions and education, the Japanese language and culture, and academic mentorship for high school students.

Skills:

- Languages: Chinese (native); English (Test of English as a Foreign Language score 113, taken November 2024).
- Programming languages: Python, C, and C++, among others.
- Tools and technologies: Git version control, LaTeX typesetting, and Structured Query Language (SQL), among others.

Community Involvement and Academic Service:

- Co-initiator, CPHOS, an academic non-profit club providing free Physics Olympiad simulations to high school students nationwide (December 2020 to present).
- Invited Online Marker, 52nd International Physics Olympiad (IPhO 2022), held in Switzerland.
- Peer Reviewer for leading international conferences in machine learning and artificial intelligence: International Conference on Learning Representations (ICLR) 2025 and 2026; Conference on Language Modeling (COLM) 2025 and 2026; and Conference on Neural Information Processing Systems (NeurIPS) 2025.